

Basic Request:

Invoke-WebRequest -Uri https://httpbin.org/get -Verbose

```
PS C:\Users\lizmo> Invoke-WebRequest -Uri https://httpbin.org/get -Verbose
VERBOSE: GET https://httpbin.org/get with 0-byte payload
VERBOSE: received 304-byte response of content type application/json

StatusCode      : 200
StatusDescription : OK
Content         : {
  "args": {},
  "headers": {
    "Host": "httpbin.org",
    "User-Agent": "Mozilla/5.0 (Windows NT; Windows NT 10.0; en-ZA)
WindowsPowerShell/5.1.19041.6456",
    "X-Amzn-Trace-Id": "Root=1-69d11...
RawContent      : HTTP/1.1 200 OK
                  Connection: keep-alive
                  Access-Control-Allow-Origin: *
                  Access-Control-Allow-Credentials: true
                  Content-Length: 304
                  Content-Type: application/json
                  Date: Sat, 04 Apr 2026 14:18:30 GM...
Forms           : {}
Headers         : {[Connection, keep-alive], [Access-Control-Allow-Origin, *], [Access-Control-Allow-Credentials,
true], [Content-Length, 304]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 304

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```

Request Headers:

```
"headers": {
  "Host": "httpbin.org",
  "User-Agent": "Mozilla/5.0 (Windows NT; Windows NT 10.0; en-ZA)
WindowsPowerShell/5.1.19041.6456",
  "X-Amzn-Trace-Id": "Root=1-69d11..."
}
```

Response Headers:

```
RawContent      : HTTP/1.1 200 OK
                  Connection: keep-alive
                  Access-Control-Allow-Origin: *
                  Access-Control-Allow-Credentials: true
                  Content-Length: 304
                  Content-Type: application/json
                  Date: Sat, 04 Apr 2026 14:18:30 GM...
```

HTTP Status code:

```
StatusCode      : 200
```

Connection Workflow:

Standard Request/Response Cycle

POST Request:

Invoke-WebRequest -Uri "https://httpbin.org/post" -Method Post -ContentType "application/x-www-form-urlencoded" -Body "key=value"

```
PS C:\Users\lizmo> Invoke-WebRequest -Uri "https://httpbin.org/post" -Method Post -ContentType "application/x-www-form-urlencoded" -Body "key=value"

StatusCode      : 200
StatusDescription : OK
Content         : {
  "args": {},
  "data": "",
  "files": {},
  "form": {
    "key": "value"
  },
  "headers": {
    "Content-Length": "9",
    "Content-Type": "application/x-www-form-urlencoded",
    "Host": "h...
RawContent      : HTTP/1.1 200 OK
                  Connection: keep-alive
                  Access-Control-Allow-Origin: *
                  Access-Control-Allow-Credentials: true
                  Content-Length: 476
                  Content-Type: application/json
                  Date: Sat, 04 Apr 2026 14:30:19 GM...
Forms           : {}
Headers         : {[Connection, keep-alive], [Access-Control-Allow-Origin, *], [Access-Control-Allow-Credentials, true], [Content-Length, 476]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 476

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```

POST vs GET:

- GET is primarily used to request or retrieve data from a server, and the parameters are usually visible in the URL.
- POST is used to send or "post" data to a server to be processed, such as submitting a form or uploading a file.

Unlike GET, a POST request includes a body (payload) that contains the data being sent.

The Purpose of the Payload (**key=value**):

- Purpose: The payload is the actual information you want the server to process. In this case, **key=value** is a simple piece of data representing a form field and its input.
- Transmission: This data is transmitted in the body of the HTTP message, not the header. Because it is in the body, it is handled separately from the URL and request metadata.

Observing the Response Body:

When you run the POST command to httpbin.org/post, the server sends back a JSON response:

```
"args": {},  
  
  "data": "",  
  
  "files": {},  
  
  "form": {  
  
    "key": "value"  
  
  },
```

Status Code Exploration:

404 (Page not found because nothing exists at that URL address)

```
PS C:\Users\lizmo> Invoke-WebRequest -Uri https://httpbin.org/status/404 -Method Head  
Invoke-WebRequest : The remote server returned an error: (404) Not Found.  
At line:1 char:1  
+ Invoke-WebRequest -Uri https://httpbin.org/status/404 -Method Head  
+ ~~~~~  
+ CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-WebRequest], WebException  
+ FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeWebRequestCommand  
  
PS C:\Users\lizmo>
```

403 (Forbidden because my client does not have authorization to access that URL address)

```
PS C:\Users\lizmo> Invoke-WebRequest -Uri https://httpbin.org/status/403 -Method Head  
Invoke-WebRequest : The remote server returned an error: (403) Forbidden.  
At line:1 char:1  
+ Invoke-WebRequest -Uri https://httpbin.org/status/403 -Method Head  
+ ~~~~~  
+ CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-WebRequest], WebException  
+ FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeWebRequestCommand  
  
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```